THICKENING INVARIANCE OF STACK-LEVEL PROPERTIES (CONRAD)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 506-524 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 526-557, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 36 of queue_stacks-stacks-more-morphisms.json (pb=1241, stmt 506-524, proof 526-557).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04X6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04SZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thickening-diagonals: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 454–468. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZR: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof

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but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 050D: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04YW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AHJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 557–557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X} \subset \mathcal{X}'$ be a thickening of algebraic stacks. Then*

(1) *$\mathcal{X}$ is an algebraic space if and only if $\mathcal{X}'$ is an algebraic space,*
(2) *$\mathcal{X}$ is a scheme if and only if $\mathcal{X}'$ is a scheme,*
(3) *$\mathcal{X}$ is DM if and only if $\mathcal{X}'$ is DM,*
(4) *$\mathcal{X}$ is quasi-DM if and only if $\mathcal{X}'$ is quasi-DM,*
(5) *$\mathcal{X}$ is separated if and only if $\mathcal{X}'$ is separated,*
(6) *$\mathcal{X}$ is quasi-separated if and only if $\mathcal{X}'$ is quasi-separated, and*
(7) *add more here.*

Proof. In each case we reduce to a question about the diagonal and then we use Lemma lemma-thickening-diagonals applied to the morphism of thickenings

$$(\mathcal{X} \subset \mathcal{X}') \to (\mathrm{Spec}(\mathbf{Z}) \subset \mathrm{Spec}(\mathbf{Z}))$$

We do this after viewing $\mathcal{X} \subset \mathcal{X}'$ as a thickening of algebraic stacks over $\mathrm{Spec}(\mathbf{Z})$ via Algebraic Stacks, Definition 04X6.

Case (1). An algebraic stack is an algebraic space if and only if its diagonal is a monomorphism, see Morphisms of Stacks, Lemma 0AHJ (this also follows immediately from Algebraic Stacks, Proposition 04SZ).

Case (2). By (1) we may assume that $\mathcal{X}$ and $\mathcal{X}'$ are algebraic spaces and then we can use More on Morphisms of Spaces, Lemma 05ZR.

Case (3) – (6). Each of these cases corresponds to a condition on the diagonal, see Morphisms of Stacks, Definitions 04YW and 050D. □